TOPICS IN SYSTEM ENGINEERING-IC DESIGN WS 2024/25

Topic 5: Arithmetic

Advanced Adder Designs for High-Performance Digital Circuits

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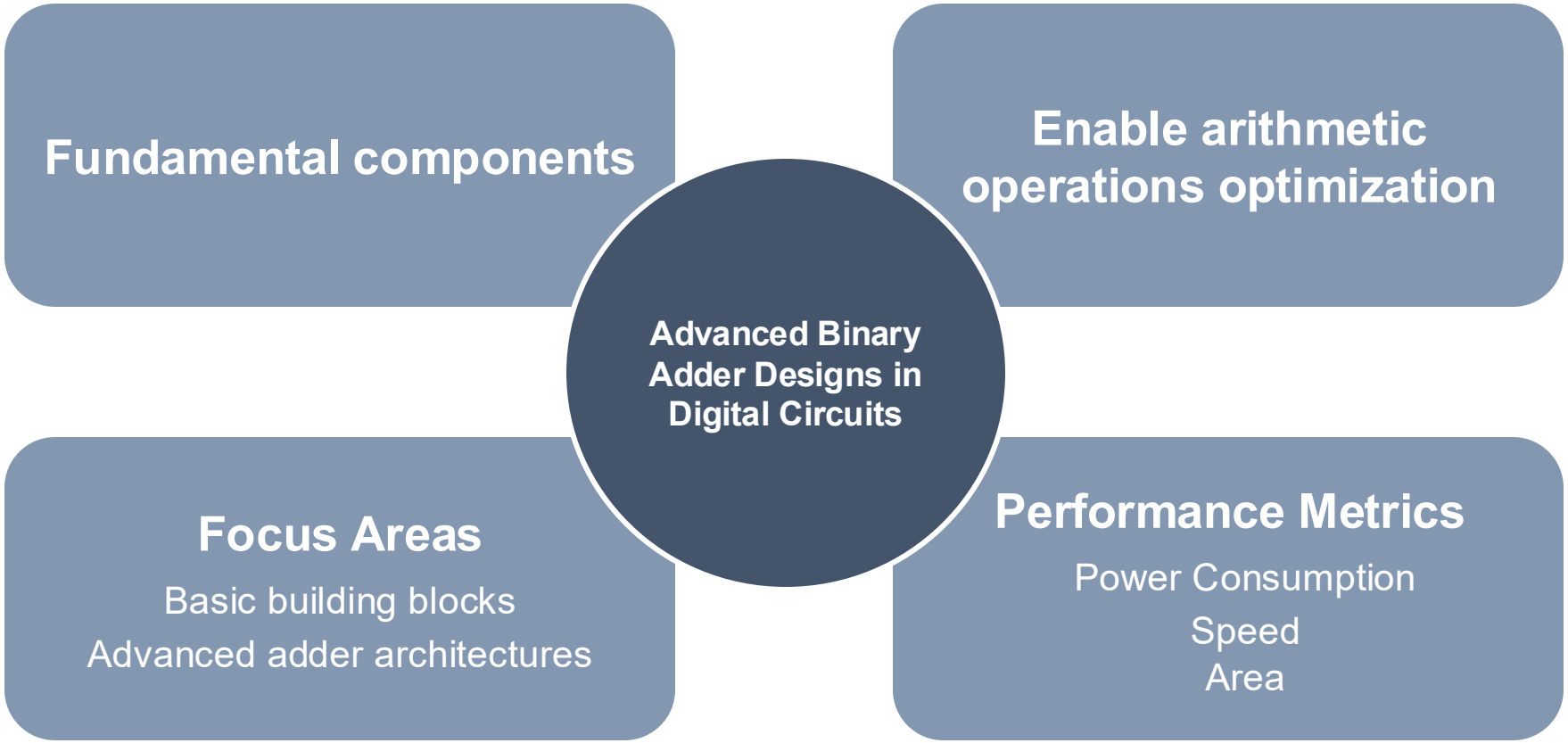
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Introduction



Motivation

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1

Challenge

Traditional adders have a delay that increases linearly with the number of bits

2

Need

High-performance applications demand efficient adders with low power and small area

3

Solution

Advanced architectures reduce critical path delay via parallel & hierarchical techniques

4

Research Areas

Balancing speed, power consumption, and area remains the core research focus

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Binary Addition

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Importance
Basis of operations in ALUs and digital processors

Process Overview
- Similar to decimal long addition but with binary digits (0 and 1)
- Involves simple sum-carry logic

Key Concepts
- LSB Addition: Adds two bits without a carry-in
- Other Bit Positions: Adds two operand bits plus a carry-in

Output Bit Width
Adding n-bit numbers typically produces an n+1 result

Table 1: Truth Table – Binary Addition

Input		Output	
A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Table 1 copied from: Abbas, Karim. “Arithmetic.” In Handbook of Digital CMOS Technology, Circuits, and Systems

CMOS

Definition	CMOS stands for Complementary Metal-Oxide-Semiconductor
Role in Digital Circuits	Fundamental technology for designing efficient logic gates
Transistor Types	➢ **pMOS**: Turn ON when the gate voltage is low ➢ **nMOS**: Turn ON when the gate voltage is high
Benefits	➢ Reliable and energy-efficient circuit design ➢ Enhanced performance and scalability ➢ Low power consumption

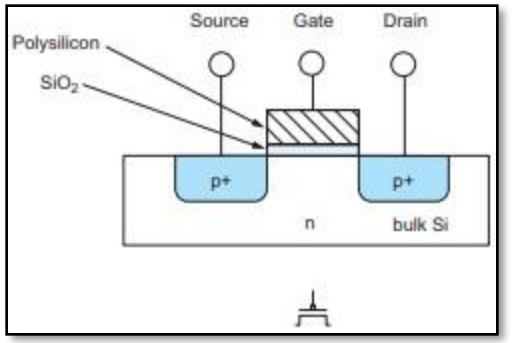


Fig 1: pMOS Transistor

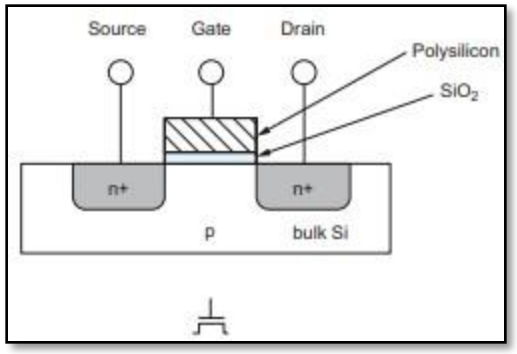


Fig 2: nMOS Transistor

Fig 1 & 2 copied from: Weste, Neil H. E., and David Money Harris. CMOS VLSI Design: A Circuits and Systems Perspective

Half Adder: Basic Building Block of Addition

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- Simplest digital circuit for adding two binary numbers
- Adds two single-bit inputs (A and B)
  - **Sum (S):** A ⊕ B (XOR operation)
  - **Carry (Cout):** A · B (AND operation)
- Limitations
  - Does not account for carry input (C_{in}) from the previous stage
  - Cannot be used for multi-bit addition

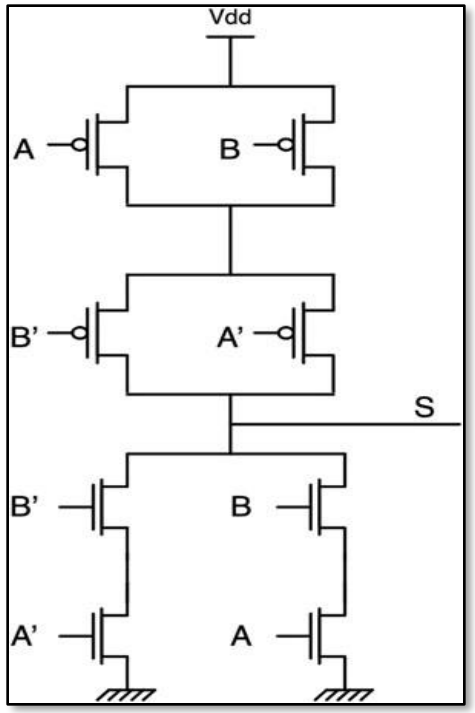


Fig 3: S gate implementation for HA

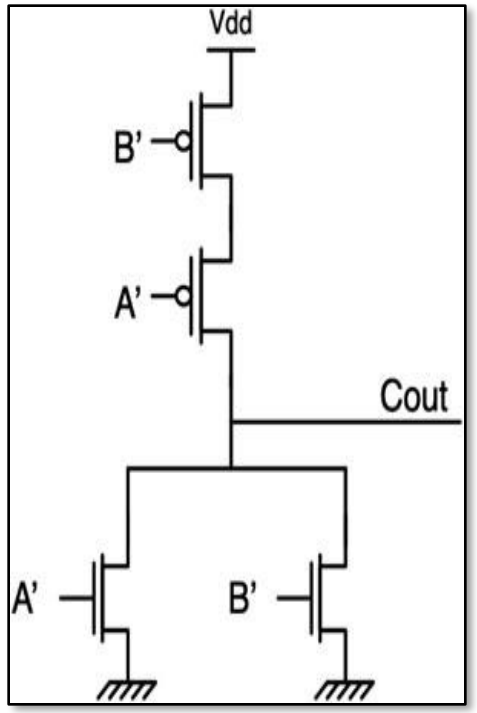


Fig 4: C_{out} gate implementation for HA

Fig 3 & 4 copied from: Abbas, Karim. “Arithmetic.” In Handbook of Digital CMOS Technology, Circuits, and Systems

Full Adder: Extending Addition Capability

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- Extends the Half Adder by including a carry-in (C_{in})
- Takes three inputs (A, B, C_{in}):
  - **Sum (S):** $A \oplus B \oplus C_{in}$
  - **Carry-out (C_{out}):** $A \cdot B + B \cdot C_{in} + A \cdot C_{in}$
- Advantages Over Half Adder
  - Can handle multi-bit binary addition by chaining
  - Generates correct carry propagation across multiple bits

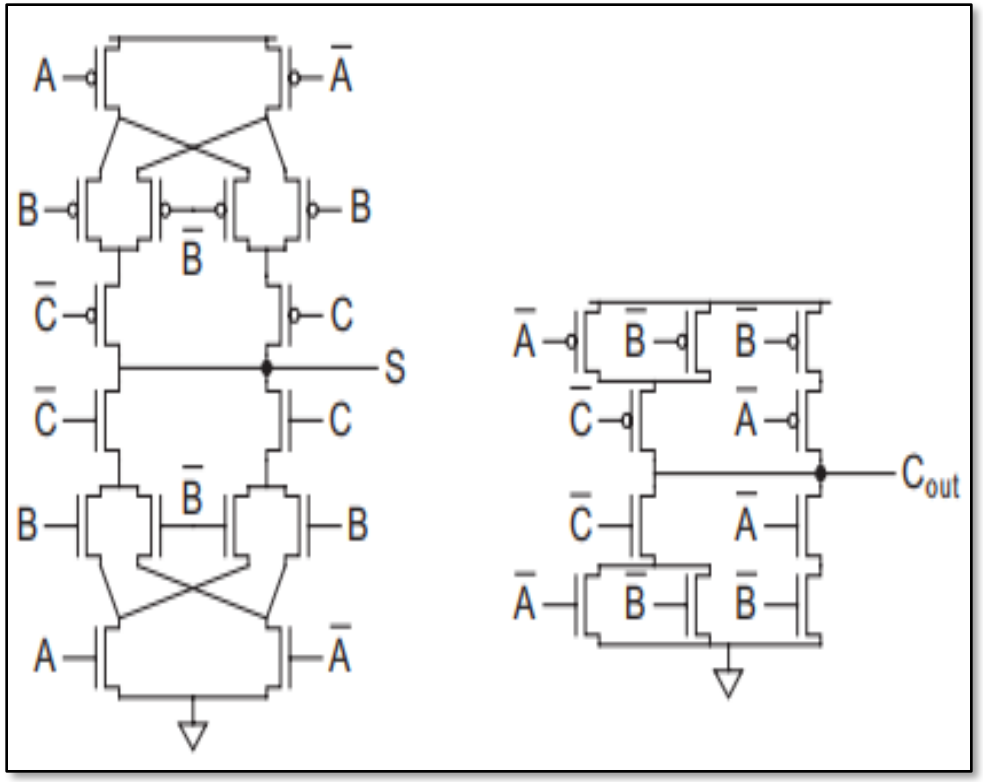


Fig 5: Sum and Cout implementation for FA

Fig 5 copied from: Weste, Neil H. E., and David Money Harris. CMOS VLSI Design: A Circuits and Systems Perspective

Ripple Carry Adder: Simple but Slow

Simplest multi-bit adder

Cascading Full Adders (FA) in sequence

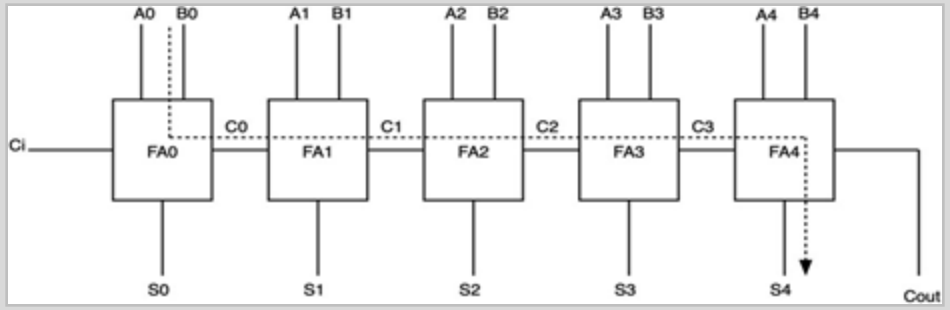


Fig 6: A 5-bit ripple carry adder (RCA)

- Each FA adds two binary numbers along with a carry-in bit
- Carry propagation
  - Carry output from one FA serves as the carry input for the next FA, creating a ripple effect
- Key characteristics
  - Simple and area-efficient (uses fewer gates)
  - Scalable for different bit widths

Fig 6 copied from: Abbas, Karim. “Arithmetic.” In Handbook of Digital CMOS Technology, Circuits, and Systems

Ripple Carry Adder: Strengths & Weaknesses

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Advantages

- Minimal hardware requirements (fewer logic gates)
- Efficient for small bit-width operations
- Simple and easy to design

Disadvantages

- Each carry bit propagates through all previous adders before the final sum is ready
- Efficient for small bit-width operations
- Slow for large bit-widths due to linear delay growth (O(N)
- Not suitable for high-speed applications like usage

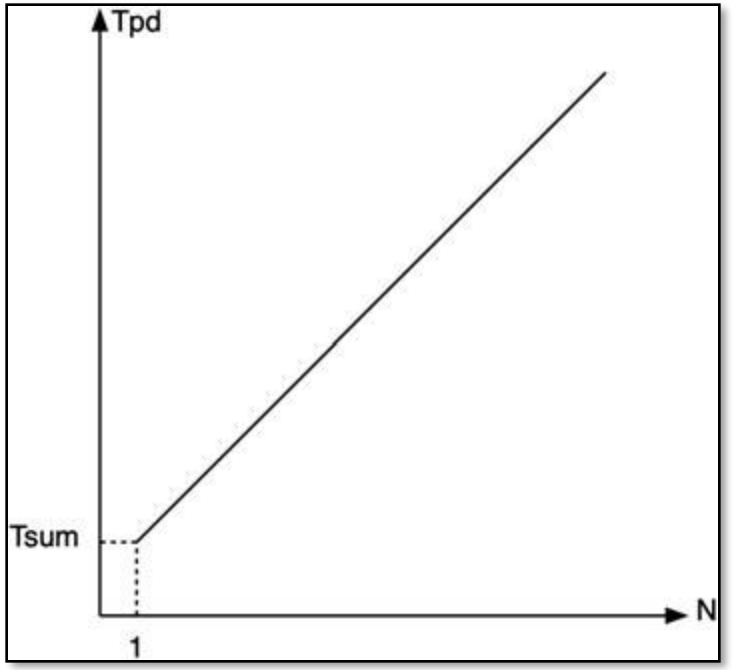


Fig 7: Path delay vs Number of Bits

Fig 7 copied from: Abbas, Karim. “Arithmetic.” In Handbook of Digital CMOS Technology, Circuits, and Systems

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Generate Propagate Logic

- Enhancement over traditional Ripple Carry Adders
- Improves carry computation:
  - By predicting carry values early in process
- Three key states in G-P Logic:
  - **Generate (G):** C_{out} is '1' regardless of C_{in}
  - **Propagate (P):** C_{out} has the same value as C_{in}
  - **Delete (D):** C_{out} is '0' regardless of C_{in}
- Helps in reducing delay and improving speed in digital addition

Table 2: Truth Table for G-P Logic

A	B	C_{in}	C_{out}	S	G	D	P
0	0	0	0	0	0	1	0
0	0	1	0	1	0	1	0
0	1	0	0	1	0	0	1
0	1	1	1	0	0	0	1
1	0	0	0	1	0	0	1
1	0	1	1	0	0	0	1
1	1	0	1	0	1	0	0
1	1	1	1	1	1	0	0

Table 2 copied from: Abbas, Karim. "Arithmetic." In Handbook of Digital CMOS Technology, Circuits, and Systems

Generate Propagate Logic

- G-P Logic equations
  - **Generate:** G = A·B
  - **Propagate:** P = A⊕B
  - **Carry-out:** $C_{out} = G+(P \cdot C_{in})$
- Implemented using Manchester Carry Chains
- Allows parallel calculation of carry bits, reducing delay
- Used in Lookahead and Carry-Bypass adders for high-speed arithmetic

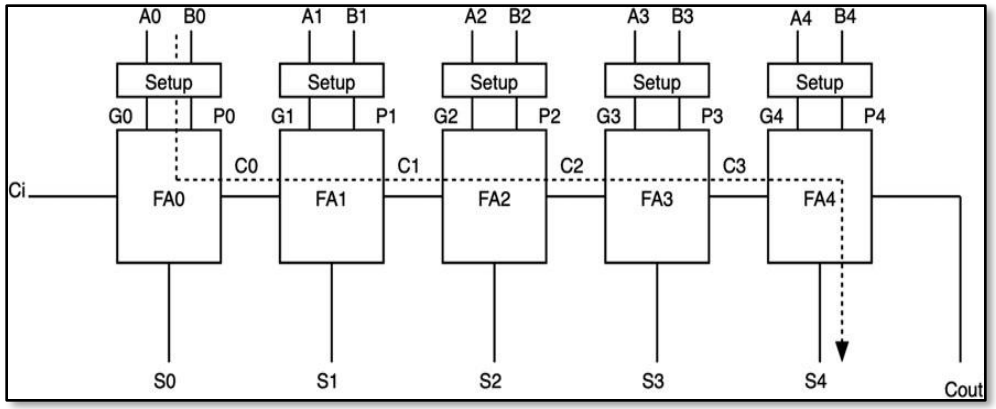


Fig 8: G-P logic implementation using ripple carry adder

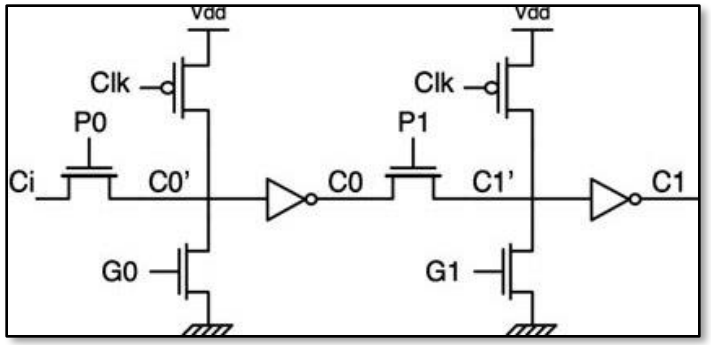


Fig 9: Manchester Carry Chain

Fig 8 & 9 copied from: Abbas, Karim. “Arithmetic.” In Handbook of Digital CMOS Technology, Circuits, and Systems

Carry Bypass Adder: Enhancing Addition Speed

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- Improve upon RCAs by reducing carry propagation delay
- No wait for the carry to ripple through all full adders
- Allows the carry to bypass certain bit positions
- Uses G-P logic to determine if the carry can be forwarded directly
- Key components:
  - Full adders with Manchester carry chain
  - Multiplexer-based bypass logic to speed up carry propagation

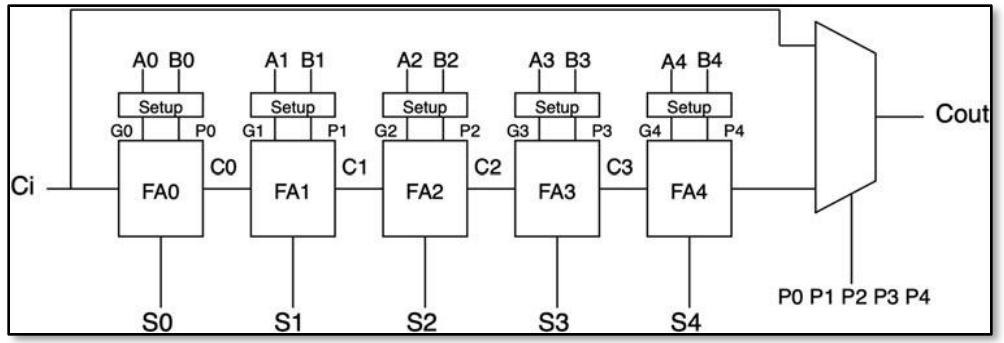


Fig 10: Modified 5-bit ripple carry adder

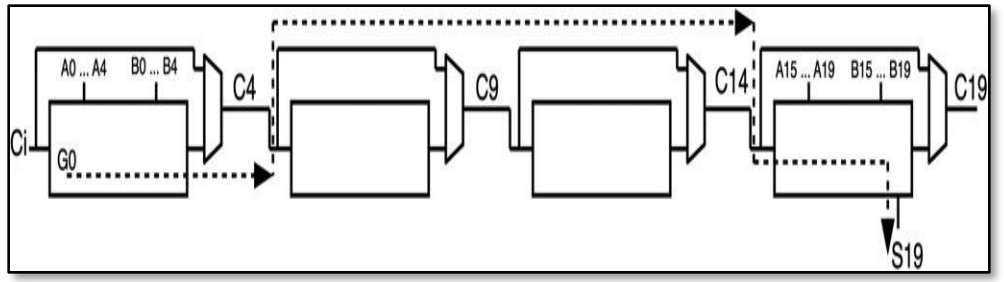


Fig 11: Critical path for 20-bit carry-bypass adder

Fig 10 & 11 copied from: Abbas, Karim. “Arithmetic.” In Handbook of Digital CMOS Technology, Circuits, and Systems

Carry Bypass Adder: Speed vs. Complexity

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Advantages

- Reduces critical path delay by enabling fast carry propagation
- Provides linear speed improvement compared to Ripple Carry Adders
- Efficient for moderate word lengths (e.g., 16-bit, 32-bit adders)

Challenges

- Requires additional multiplexers, increasing circuit area
- Performance gain depends on optimal block size selection
- Not as fast as Carry-Lookahead Adders for very large bit widths

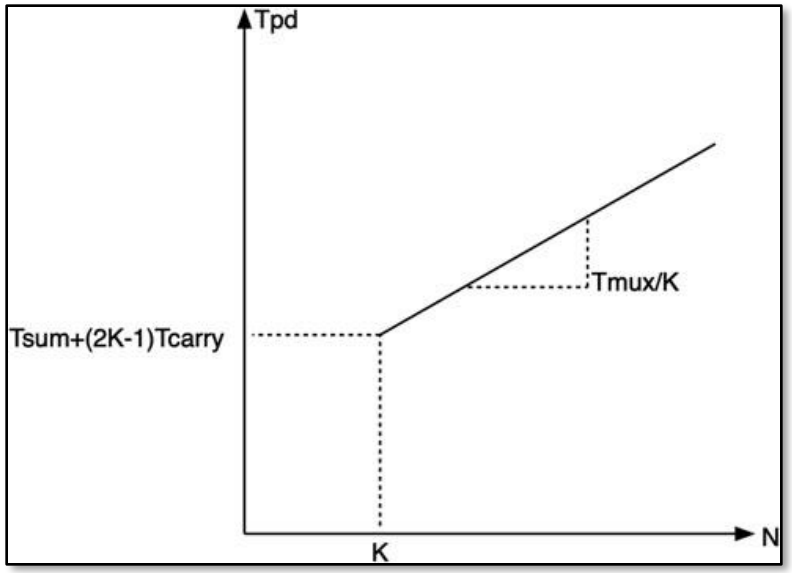


Fig 12: Carry-Bypass delay with N

Fig 12 copied from: Abbas, Karim. “Arithmetic.” In Handbook of Digital CMOS Technology, Circuits, and Systems

Carry-Save Adder: Faster Multi-Operand Addition

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- Improves speed in multi-operand addition by avoiding immediate carry propagation
- CSA stores carry bits separately, reducing overall computation time.
- Uses multiple full adders (FA) in parallel
- No Immediate carry propagation, saved for a later stage
- Instead of producing a final sum and carry immediately, two separate values are generated:
  - **Sum (S)** – Stores the partial sum
  - **Carry (C)** – Stores carry bits, which are later added
- Benefits:
  - Faster than Ripple Carry Adder for multi-operand addition
  - Reduces critical path delay, suitable in high-performance usage

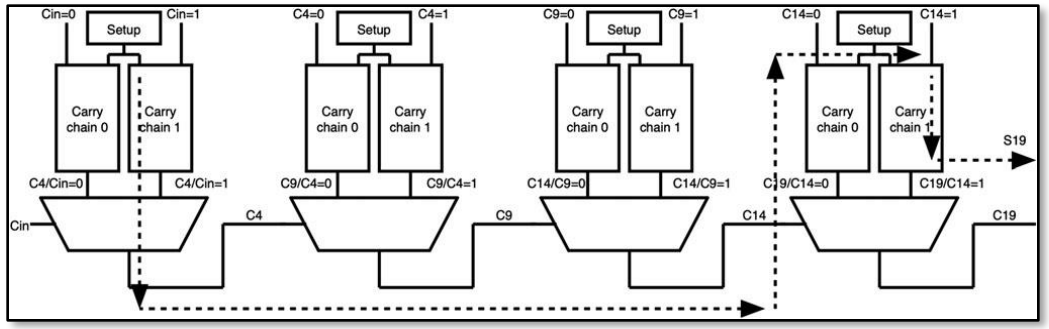


Fig 13: Carry Save adder design

Fig 13 copied from: Abbas, Karim. “Arithmetic.” In Handbook of Digital CMOS Technology, Circuits, and Systems

Lookahead Adder: A High-Speed Adder Design

- RCAs suffer from linear delay dependence on word length (O(N))
- Lookahead addition reduce delay by computing carries in advance
- No sequential propagation of carries
- Uses G-P logic to calculate carry bits in parallel
- Aims to make the addition delay independent of bit count
- Key benefit:
  - Faster addition by eliminating ripple effect in carry propagation.
- Trade-off:
  - Increased circuit complexity and power consumption

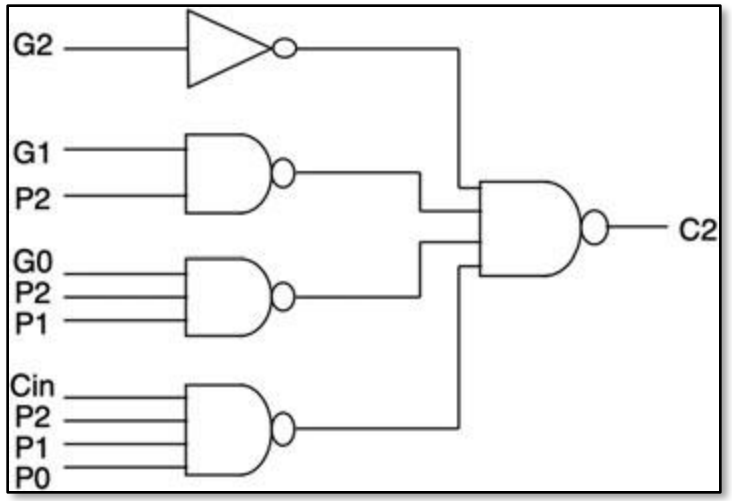


Fig 14: Lookahead logic

Fig 14 copied from: Abbas, Karim. “Arithmetic.” In Handbook of Digital CMOS Technology, Circuits, and Systems

Lookahead Adder: A High-Speed Adder Design

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- Carry Calculation Formula
  - Carry is computed using: $C_n = G_n + (P_n \cdot C_{n-1})$
  - Expanding for multiple bits:

    $C_n = G_n + (P_n \cdot G_{n-1}) + (P_n \cdot P_{n-1} \cdot G_{n-2}) + \ldots$
- Instead of waiting for each carry bit to propagate, all carry values are computed at once
- Circuit consists of:
  - Generate and Propagate Logic (to precompute carry values)
  - Lookahead Carry Unit (which calculates carry bits in parallel)
- Performance Impact
  - Significantly reduces critical path delay
  - Works well for small-to-medium word lengths
  - Becomes inefficient for extremely large word sizes

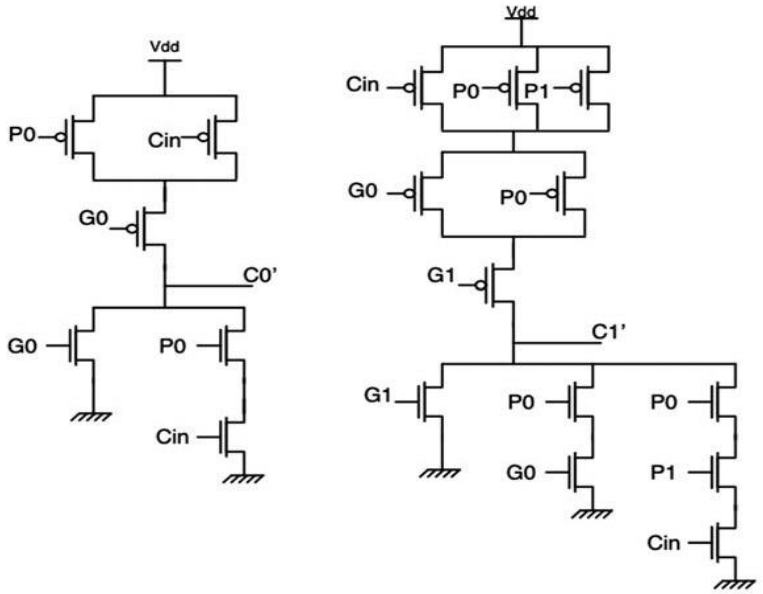


Fig 15: Lookahead Adder for bit position 1 and 2

Fig 15 copied from: Abbas, Karim. “Arithmetic.” In Handbook of Digital CMOS Technology, Circuits, and Systems

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Summary & Outlook

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Summary

- Adder circuits are essential for high-performance digital computing
- Ripple Carry Adders are simple but suffer from high delay
- Advanced adders (Carry-Save, Carry-Bypass, Lookahead) optimize speed and efficiency
- Trade-offs exist between speed, power consumption, and circuit complexity

Outlook

- Energy-Efficient Designs: Focus on low-power adders for battery-operated devices
- Scalability: Adder architectures optimized for AI, ML, and high-performance computing
- Exploration of non-traditional adder designs for next-gen processors
- Combining multiple techniques to achieve optimal performance across diverse applications

References

[1] Abbas, Karim. “Arithmetic.” In Handbook of Digital CMOS Technology, Circuits, and Systems, edited by Karim Abbas, 439–51. Cham: Springer International Publishing, 2020. https://doi.org/10.1007/978-3-030-37195-1 11.

[2]Weste, Neil H. E., and David Money Harris. CMOS VLSI Design: A Circuits and Systems Perspective. 4th ed. Boston: Addison Wesley, 24-88, 2011.

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